

Algebra PhD Exam, September 1990

Read through the questions carefully before you begin. Your six best solutions will count. Theorems from the course may be quoted (within reason) as long as you give a clear indication and state the result in full.

1. Let P be a nontrivial, finite p -group, and Q a subgroup. Consider the action of P on the set P/Q of left cosets of Q by left multiplication:

$$P \times P/Q \longrightarrow P/Q, \quad (p, xQ) \longmapsto (px)Q.$$

Q acts on P/Q by restriction.

- a) Show that xQ is a fixed point of Q if and only if $x \in N_P(Q) = \{p \in P \mid pQp^{-1} = Q\}$.
 - b) Deduce that if Q is a proper subgroup of P then Q is also a proper subgroup of $N_P(Q)$.
 - c) Hence show that every maximal subgroup of P is normal of index p .
2. Let A be a Dedekind domain with field of fractions K . Show that every ideal of A can be generated by two elements. (Hint: use the Chinese remainder theorem.)
 3. Prove that any simple group of order 60 is isomorphic to A_5 . (you need not show that A_5 is simple).
 4. Let A, B and C be rings (with 1) and suppose we have a commutative diagram of ring homomorphisms:

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & B \\ & \searrow \gamma \quad \swarrow \beta & \\ & C & \end{array}$$

- a) Define the structure of an (A, B) bimodule on B , ie make B into a left A -module and a right B -module such that $(a \cdot x)b = a(x \cdot b)$ for $a \in A, x, b \in B$.
- b) Show that for any right A -module M , we have a natural isomorphism

$$M \otimes_A C \cong (M \otimes_A B) \otimes_B C.$$

(Remember what ‘natural’ means in terms of the functors $- \otimes_A C$ and $(- \otimes_A B) \otimes_B C$.)

5. Let $\{\sigma_i\}_{i=1}^n$ be a set of n distinct automorphisms of a field k . Show that if for $a_1, \dots, a_n \in k$,

$$\sum a_i \sigma_i(u) = 0 \quad \text{for all } u \in k,$$

then $a_i = 0$ for all i . (Hint: show that there is no ‘shortest’ relation with $a_i \neq 0$.)

6. Give examples of the following (with proof).
 - a) An irreducible cubic polynomial with rational coefficients whose Galois group has order 3.
 - b) Two normal extensions $K \subset L, L \subset M$ such that extension $K \subset M$ is not normal.
7. Let R be a ring and M simple (irreducible) R -module.
 - a) Prove Schur’s Lemma: every nonzero R -module map from M to itself is an isomorphism.
 - b) Give an example to show that the converse of Schur’s Lemma is false.

8. Let $\mathbb{Z}_p$ denote the p -adic integers and for $n \in \mathbb{N}$,

$$\varepsilon_n : \mathbb{Z}_p \longrightarrow \mathbb{Z}/p^n\mathbb{Z},$$

the map defining the structure of $\mathbb{Z}_p$ as $\varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$. Let $U_n = 1 + p^n\mathbb{Z}_p$.

- a) Show that U_n is a subgroup of the group U of units of $\mathbb{Z}_p$.
- b) Show that U_n is the kernel of $\varepsilon_n|_U : U \rightarrow (\mathbb{Z}/p^n\mathbb{Z})^*$ and that $U/U_1 \cong \mathbb{F}_p^*$ is cyclic of order $p-1$.
- c) For $n \geq 1$ show that the map

$$1 + p^n x \mapsto x \text{ modulo } p$$

defines an isomorphism

$$U_n/U_{n+1} \xrightarrow{\sim} \mathbb{Z}/p\mathbb{Z}.$$

- d) Deduce that U_1/U_n is a p -group and that U/U_n has a unique subgroup of order $p-1$.
- e) Prove that U has a subgroup of order $p-1$. (Hint: $U \cong \varprojlim U/U_n$.)

9. Let D be a division ring and M a D -module. A subring R of $\text{End}_D(M)$ is said to be dense if for any natural number n and $v_1, \dots, v_n \in M$ which are linearly independent over D , and any n elements $w_1, \dots, w_n$ of M , there exists $r \in R$ such that $w_i = r(v_i)$, $i = 1, \dots, n$.

- a) Show that if $\dim_D M$ is finite then a dense subring of $\text{End}_D M$ must be equal to $\text{End}_D M$.
- b) Show that if M is infinite-dimensional over D and R is a dense subring of $\text{End}_D M$, then for every natural number m , R has a subring S_m which maps homomorphically onto $\text{Mat}_m(D)$.

10. This problem has three parts.

- a) State Hilbert's Nullstellensatz, giving the main definitions involved.
- b) Let k be an algebraically closed field. Show that every algebraic subset of k^n can be decomposed uniquely as a finite union of (irreducible) varieties.
- c) Give an example to show that in b) the union may not be disjoint.